

# **From One-Off Activity to Teaching Sequence**

## **Seminar 3 • Playful Programming and Artificial Intelligence**

University of Copenhagen • Department of Computer Science

# The problem with great workshop moments

You have all seen it — perhaps run it:

A brilliant activity. Students engaged, laughing, arguing.

Two weeks later: *"Wait, did we do something about AI?"*

**Engagement is not retention. A moment is not a sequence.**

This hour: what research says about the difference — while you are mid-design, when it can still change what you build.

# Why one-offs fade: three findings

1. **Isolated episodes don't transfer.** A concept met once, in one context, stays welded to that context — the classifier is "the sticky-note game", not an idea about data.
2. **Novelty masks learning.** High engagement in a single session often reflects the *newness*, not the concept. Strip the novelty and test what remains.
3. **No return, no depth.** Understanding grows by *revisiting* an idea in a new form — not by meeting it once vividly.

The fix is not better one-offs. It is **sequences**.

# What a sequence is (and is not)

**Not:** the same activity repeated.

**Not:** a topic block ("three weeks on AI").

**A sequence is one core question, revisited through escalating encounters:**

unplugged → plugged → applied → critiqued

Each encounter strips a simplification from the last —  
transposition, staged over time.

# You already know one sequence: this workshop

| Encounter                | Form           | What was removed                     |
|--------------------------|----------------|--------------------------------------|
| "Is this AI?"            | argument       | nothing — pure intuition             |
| Sticky-note trees        | paper model    | vagueness: now there's a mechanism   |
| Storytelling game        | embodied model | determinism: now there's probability |
| Sensors + LLM bridge     | real system    | pretence: now it's the actual thing  |
| Lying, confident machine | critique       | trust: now you interrogate it        |

**Day 1 to Day 2 was a designed sequence.** Steal the structure, not the activities.

# The study-and-research angle

The Danish/French didactics tradition offers a sharper version:

Start not from content but from **a question worth investigating** —

*"Can our classroom tell who is in it?"*

*"Why does the chatbot speak worse Danish than English?"*

*"What does the school building know about us?"*

A good question **generates** the sequence: students need the concepts to pursue the answer. Need precedes explanation.

# Anatomy of a sequence question

Test your driving question against three criteria:

1. **It requires the concept** — you cannot answer it without meeting classification, prediction, or data honestly
2. **It survives an answer** — the first answer opens two more questions
3. **It belongs to the students** — situated in their room, their language, their data

*"What is machine learning?"* fails all three.

*"Can the AI tell when our class is bored?"* passes all three — and smuggles in ethics without announcing it.

# Practical patterns for gymnasium reality

You do not control the timetable. Sequences that survive contact with a Danish gymnasium:

- **The 4 × 20-minute thread** — one encounter per week inside existing lessons; beats a single 90-minute event
- **The returning artefact** — the paper tree built in week 1 is *retested* in week 3 against a real model
- **The cross-subject relay** — the same question passes from biology (the data) to Danish (the language of the output) to samfunds-fag (who is accountable)
- **The October move** — end the sequence by having students redesign the first activity for a younger class. Transposition as assessment.